Introduction to ORCAD

(EBN 111 : Electricity and Electronics)

Pre-Practicum 1

Department of Electrical, Electronic and Computer Engineering

UNIVERSITEIT VAN PRETORIA
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EBN 111: Practical 1 Objectives

- Familiarize students with OrCAD
- Execute circuit simulation in OrCAD
- Complete Practicum Assessment
  - Practical 1 will take place during the week of **4 – 8 March 2013**
  - Group allocations will be posted on clickUp

EBN 111: Practical 1
Practical groups

Day	Time	Discipline to attend
Monday	11:30-14:20	Mechanical 1st Year (Surnames: Agigi-Maswanganyi)
Monday	14:30-17:20	Chemical (All), Civil (All), Electrical (All), Industrial 2nd Year, Mechanical 2nd Year
Wednesday	10:30-13:20	Mining (All), Industrial 1st Year (Afrikaans group), Metalurgical 2nd Year, Electronic (All),
Wednesday	14:30-17:20	Computer (All), Industrial 1st Year (English group)
Friday	13:30-16:20	Mechanical 1st Year (Surnames: Matheba-Zackarias)

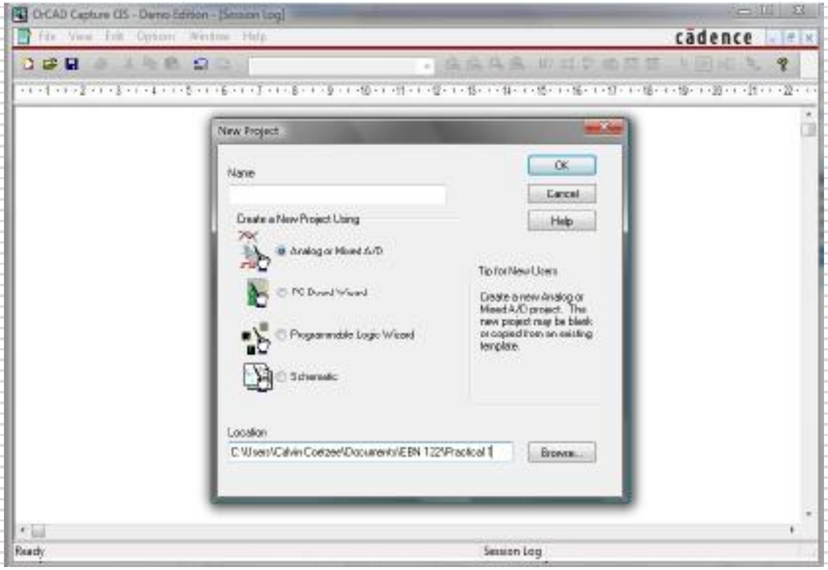
EBN 111: Practical 1
OrCAD

- Schematic Capture and circuit Simulation are two essential tools in computer aided circuit analysis and design
- Computer-aided design makes it possible to deal with problems which are computationally very complex and would be lengthy without the use of a computer
- Simulation software is part of modern engineering practice
- This practicum provides an opportunity to gain some skills in using these tools, which are used in many subsequent modules

EBN 111: Practical 1
OrCAD

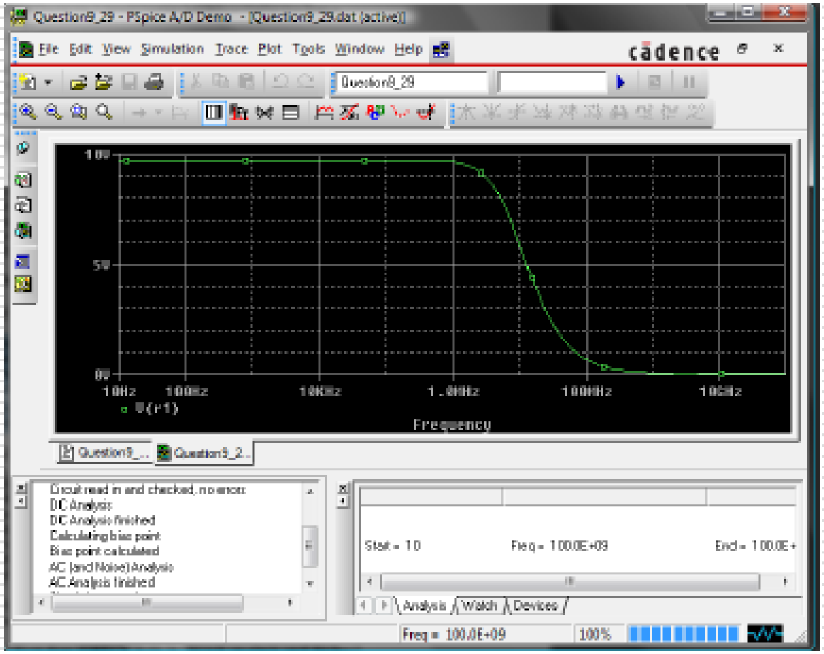
- What is Schematic Capture?

- Schematic Capture is an electronic means of graphically constructing a circuit in a format that can be subsequently read by software such as PSpice for circuit simulation and by OrCAD Layout for circuit board layout



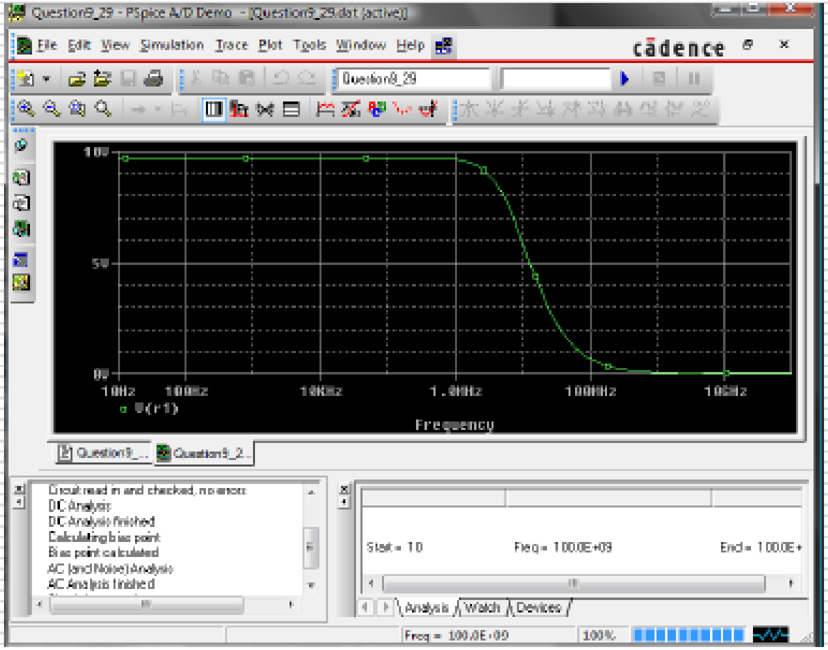
EBN 111: Practical 1 OrCAD

- What is SPICE and what is PSpice?
- SPICE was originally an analogue circuit simulator software package that was developed at the University of California at Berkeley
- PSpice is a version of SPICE for personal computers



EBN 111: Practical 1 OrCAD

- How does PSpice help in circuit analysis and design?
- PSpice's strong point is that it helps the user simulate and analyse the circuit design graphically on the computer without having to build a physical circuit or perform tedious calculations by hand



EBN 111: Practical 1 Preparation

- Log on to ClickUP and go to EBN111's page
- Click on the Practicals tab and select Practical 1
- Open the Practical 1 – Instructions.pdf file
- This file summarises what is required for practical 1.
- Perform all the tasks described in the Practical 1 – Instructions.pdf document

EBN 111: Practical 1
Practicum requirements

- What is required in order to complete the practical?
  - Complete the tutorial
  - Complete the pre-practical assignment
  - Attend the 3 hour practical session
  - Complete the pre-prac test
  - Complete the practical assignment on ClickUP during the 3 hours

EBN 111: Practical 1 Tutorial

On clickUp you will find a tutorial which describes how to use OrCAD. This is optional but it is recommended that you complete this.

Also on clickUp you will find your pre-prac assignment. This will not be evaluated, but you will not be allowed into the practical venue unless you have completed it.

EBN 111: Practical 1
Pre-practical Assignment

- The Pre-practical assignment can be found on ClickUP
- After reading through the slides, completing the tutorial and the pre-practical assignment, you should be sufficiently prepared for the practical session

EBN 111: Practical 1

Demonstration of different simulations

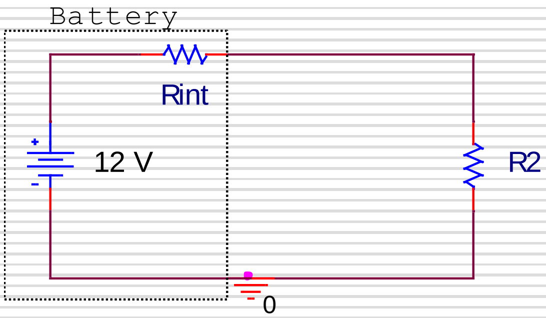
- A demonstration is given in class:
  - how to complete different types of simulations, and
  - how to view and analyse the results
- You can download a student version of the software for working at home
- The OrCAD software is available in any of the engineering laboratories

EBN 111: Practical 1
The Practical Test

- At the start of the practical session – 30 minute test.
- Each student completes test individually
- Questions will test content mentioned during today’s lecture, these slides, the tutorial, the pre-prac assignment and your circuits knowledge up to now

EBN 111: Practical 1 Example Question

- Consider the figure below
  - The battery has **internal resistance** of 5Ω
  - R2 is the **load resistance**
  - R2 must be varied sequentially from 1– 10Ω



- Complete a simulation of the circuit for R2 = 5Ω
  - Determine the **current** through R2
  - Determine the **voltage** across R2

- Sequentially set the value of R2 to five values in the range 1 - 10Ω
  - Determine the **load voltage** (V2) across R2 and the **current** for each resistor value

EBN 111: Practical 1 Practical

- After the test, you will work in groups of two to complete the practical

PRINT PRACTICAL 1 GUIDELINES (found on ClickUP) AND BRING YOUR PRINTOUT TO THE PRACTICAL SESSION!

Use the guidelines to complete the practical and then after completion – log on to ClickUP, start Practical 1 Online Evaluation and fill in your answers

EBN 111: Practical 1
Practical Assessment - Online

- Practical assessment breakdown
  - Pre-Practicum Test: 50 %
  - Practical: 50 %

EBN 111: Practical 1 Conclusion - Enjoy

- **Contact details for Questions or Feedback**
  - *Email:* u29009457@tuks.co.za
    or rethamari@gmail.com
    - **NB:** For all emails use the subject "**[EBN 111]** *subject*"
  - *Email to make appointment*

- **During practical**
  - There will be four Demis to assist you during the whole duration of the practical
  - Ask questions and remember you can also refer to the practical guide or materials available on the module website